Avoiding Layoffs: On-the-job Search and Partial Insurance

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Introduction

- Consumption smoothing over time and states-of-nature has been a perennial theme in Economics.
- Continuum from Arrow and Debreu (1654); McKenzie(1954) markets to incomplete markets with usually one risk-free asset Hall and Mishkin (1982); Zeldes (1989); Aiyagari (1994)
- Smoothing possibilities always linked to assets, rarely other mechanisms are considered.
- In general, individuals can exert effort to prevent bad shocks in the future.
- We try to explain three empirical facts:
  - (i) About 20% of job-to-job movers take wage cuts.
  - (ii) Employed job seekers are much more likely to be unemployed in the future.
  - (iii) Limited asset participation in asset markets.

A simple model

- Time is discrete. Agents maximize the present discounted value of utility flows given by:

$$\sum_{t=0}^{\infty} \beta^t (u(c_t) - \lambda s)$$

- where $u(c)$ is the utility from consumption and λs is the utility cost of exerting effort s.
- Agents exerting effort s receive offers drawn from cdf $F(w)$ with probability $p(s)$. If they accept, they get paid w each period. Otherwise, they get the wage from previous job if employed, or an employment benefit income b if unemployed.
- Agents are subject to idiosyncratic dismissal shocks. Let x be the underlying variable that determines the dismissal, and follows an AR(1) process

$$x_{t+1} = (1-\rho)\bar{x} + \rho x_t + \varepsilon_{t+1}$$

Agents with idiosyncratic shock x have a probability $\delta(x)$ of being dismissed. The value function of being unemployed is

$$U = \max_s u(b) - \lambda s + \beta\left\{p(s)\int_{\underline{w}}^{\bar{w}} \max\{W(w,\bar{x}), U\}dF(w) + (1-p(s))U\right\} \tag{1}$$

The value function of being employed is

$$W(w,x) = \max_s u(w) - \lambda s + \beta\left\{p(s)\int_{\underline{w}}^{\bar{w}} \max\{W(w',\bar{x}), \int W(w,x')dG(x'|x)\}dF(w) + (1-p(s))\left(\delta(x)U + (1-\delta(x))\int W(w,x')dG(x'|x)\right)\right\} \tag{2}$$

Optimal effort is determined by

$$\lambda = \beta p'(s)\int_{\underline{w}}^{\bar{w}} \max\{W(w,\bar{x}) - U, 0\}dF(w) \tag{3}$$

For the employed worker, the optimal effort is determined by

$$\lambda = \beta p'(s)\Bigg\{\underbrace{\int_{\underline{w}}^{\bar{w}} \max\{W(w',x) - \int W(w,x')dG(x'|x), 0\}dF(w')}_{\text{gain from switching jobs}} + \underbrace{\delta(x)\left(\int W(w,x')dG(x'|x) - U\right)}_{\text{gain from avoiding getting fired}}\Bigg\} \tag{4}$$

Full model

We extend the model to allow for capital markets. Let a be the amount of wealth the agent holds. There is a no borrowing constraint $a \geq 0$. Consistent with the evidence provided by Faberman, Mueller, Şahin, and Topa (2017) we allow differences in the wage offer distribution for unemployed workers $F_U(w)$ and employed workers $F_W(w)$

The value function of being unemployed is

$$U(a) = \max_{a',s} u(a(1+r) - a' + b) - \lambda s + \beta\left\{p(s)\int_{\underline{w}}^{\bar{w}} \max\{W(w,\bar{x},a'), U(a')\}dF_U(w) + (1-p(s))U(a')\right\} \tag{5}$$

The value function of being employed is

$$W(w,x,a) = \max_{a',s} u((1+r)a - a' + w) - \lambda s + \beta\Bigg\{p(s)\int_{\underline{w}}^{\bar{w}} \max\{W(w',\bar{x},a'), \int W(w,x',a')dG(x'|x)\}dF_W(w) + (1-p(s))\left(\delta(x)U(a') + (1-\delta(x))\int W(w,x',a')dG(x'|x)\right)\Bigg\} \tag{6}$$

The optimal effort is determined by equations similar to (3) and (4). The consumption-saving decisions is determined by the Euler equation

$$u_1((1+r)a - a' + y_0) = \beta(1+r)\mathbb{E}[u_1((1+r)a - a' + y)|s^*, a'] + \mu \tag{7}$$

where μ is the Lagrange multiplier of the borrowing constraint and y_0 is the current income, either b or w depending on employment status.

Equation (7) shows that the optimal effort decision is affected by the amount of assets agents hold. By exerting effort, agents are effectively changing the probability distribution over income next period.

Empirical Evidence

We use data from the Survey of Consumer Expectations (SCE) of the New York Fed and its Labor Job Search (JS) Supplements.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Lag inv Layoff pr (x_{t-1})	0.651***	0.615***	0.650***	0.613***	0.652***	0.625***	0.652***	0.624***
	(0.015)	(0.051)	(0.015)	(0.051)	(0.015)	(0.049)	(0.015)	(0.049)
Known lh wage		-0.012		-0.014		-0.034**		-0.035**
		(0.014)		(0.015)		(0.014)		(0.015)
Observations	32,427	2,444	32,427	2,444	32,147	2,400	32,147	2,400
R-squared	0.402	0.351	0.402	0.353	0.406	0.379	0.406	0.381
Year & Region FE	-	-	✓	✓	-	-	✓	✓
Demographics	-	-	-	-	✓	✓	✓	✓

Note: Robust standard errors in parentheses . *** $p < 0.01$, **$p < 0.05$, * $p < 0.1$

Table 1: Dynamics of Inverse of Layoff probability

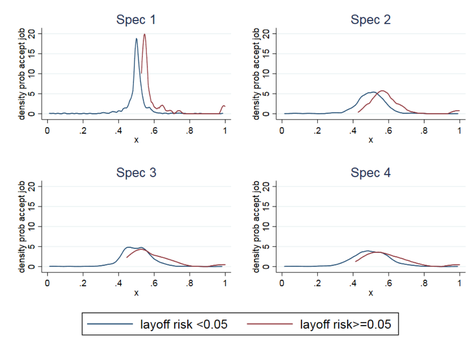
Variables	N^o methods	Hours search	N^o applications
	Year, Region FE & Demographics		
	(10)	(11)	(12)
Layoff prob	2.656***	2.395	1.846**
	(0.718)	(1.612)	(0.823)
Observations	1,988	1,988	2,491
R-squared	0.185	0.082	0.005

Note: Robust standard errors in parentheses . *** $p < 0.01$, **$p < 0.05$, * $p < 0.1$

Table 2: Expected search effort as a function of Layoff probability

	(1)	(2)	(3)	(4)
log wage gap (offered - current)	0.377***	0.388***	0.372***	0.385***
	(0.111)	(0.110)	(0.115)	(0.114)
layoff prob	2.312*	2.424*	2.587*	2.673*
	(1.277)	(1.323)	(1.447)	(1.478)
log wage gap X layoff prob	-5.635	-6.590*	-6.587	-7.451*
	(3.774)	(3.906)	(4.138)	(4.243)
Observations	570	570	561	561
Year & Region FE	-	✓	-	✓
Demographics	-	-	✓	✓

Table 3: Estimates for acceptance of job offer



Calibration

Parameter	Meaning	Value
β	Discount factor	$(0.95)^{1/52}$
λ	Cost of exerting effort	0.01
σ	Risk aversion	2
r	Interest rate	$(1.03)^{1/52} - 1\%$
b	Unemployment benefits	10 percentile of u distribution
ρ	Persistence of dismissal shock	0.707
σ_ε	Std of dismissal shock	$(0.619)^{1/2}$
F_u, F_w	Hourly wage distribution	Log normal
$(\mu_u, \sigma_u), (\mu_e, \sigma_e)$	mean and std of u and e distributions	$(2.639, 0.35), (2.796, 0.4)$
p	job finding rate as function of effort	Exponential distribution
μ	parameter exp. dist.	1

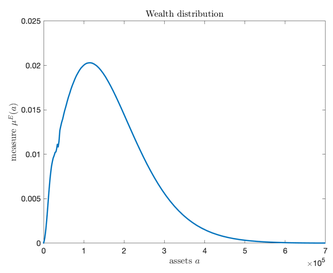
Table 4: Calibration

Model Fit

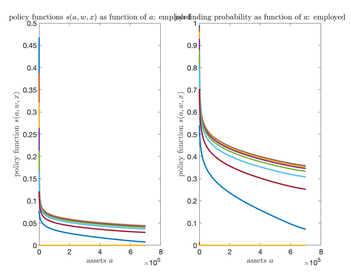
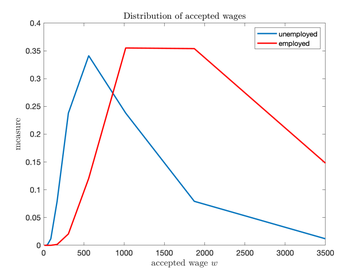
Moments	Value
Unemployment rate	5.04%
Share of households with zero wealth	0.1%
EU flows	2.23%
UE flows	2.23%
EE flows	5.2%
Mean unemployment duration	2.29 weeks

Table 5: Moments

The wealth distribution



Optimal Search Effort



Dist of accepted wages